EEES302 - Digital Design
Experiment 1 Logic Gates and Boolean Algebra

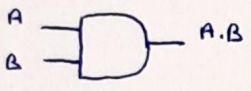
Objective:

To build a circuit from a truth table. Determining the Boolean expression for given or constructed table. Then the circuit is built from AND gates that gives the outputs into multiple input OR gate.

Summary:

The experiment includes two Boolean expression that simplified. These two expressions are written into truth table. By using truth table the circuit is constructed and output values of functions are measured by using oscillator.

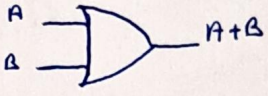
2-input AND gate



A	B	F
0	0	0
0	1	0
1	0	0
1	1	1

F= A.B

2-input OR gate

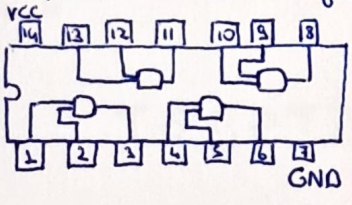


A	B	F
0	0	0
0	1	1
1	0	1
1	1	1

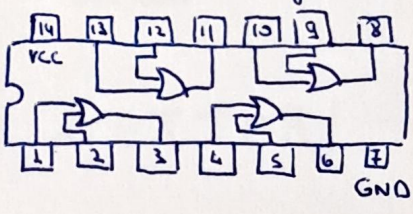
F=A+B

Equipment List:

74LS08 TTL AND gate IC



74LS32 TTL OR gate IC

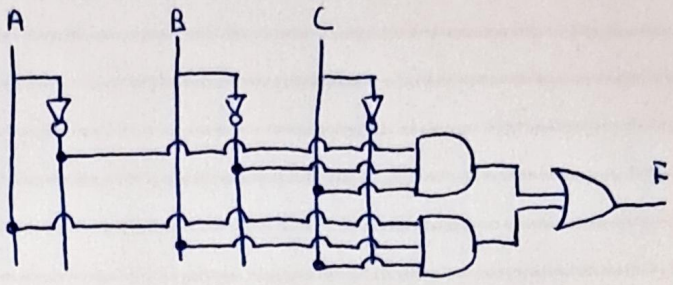


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Preliminary Work:

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1



$F = A'.B'.C + ABC$

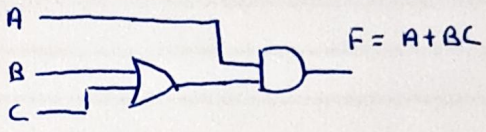
Experimental Procedure:

Equation 1: $F = A + BC + AC = A(1+C) + BC = A + BC$

Truth Table of equation:

A	B	C	A+BC
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Circuit Implementation:



Equation 2: $F = A + AB = A(1+B) = A$

Truth Table of equation:

A	F=A
0	0
1	1

Circuit Implementation:

A ——— (Line)

Result:

The equations are simplified, then the circuits are built. The working principle of AND and OR gate is understood by testing circuits. Logic high and Low inputs are given and we measured the output of the circuits. The truth table and measured values are compared. The experiment gived desired output voltages from the truth tables.

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